

Roadmap: From 2D→3DGS to Immersive Anime Reality

Vision: Building a Sword Art Online-like Interactive Anime World

Date: 2026-01-07 Current State: Velowen.art (2D Image → 3D Gaussian Splatting) Dream Goal: Fully immersive, interactive anime world with real-time navigation and character interaction

Executive Summary

Your dream of creating an interactive anime world like Sword Art Online requires bridging **5 critical technology gaps**:

1. **Static → Dynamic**: From frozen 3D scenes to animated, living worlds
2. **Single Object → Full World**: From isolated conversions to unbounded environments
3. **View-Only → Interactive**: From passive observation to active participation
4. **Isolated → Persistent**: From one-off generations to coherent, evolving worlds
5. **Silent → Social**: From empty scenes to AI-driven characters and NPCs

Good News: Your research collection contains **all the key technologies** needed. The path is clear.

Phase 1: Foundation Enhancement (Current → 3 months)

Goal: Make your 3DGS system production-ready and scalable

Technologies from Your Research:

1.1 Quality Improvements

- **Mip-Splatting** (Readiness: 9/10)
- **What:** Anti-aliasing for 3DGS via mip-map filtering
- **Why:** Eliminates artifacts when users zoom/move camera
- **Implementation:** Fork mip-splatting repo, integrate 3D smoothing + 2D Mip filters into your pipeline
- **Impact:** Crisp anime lines without shimmering
- **Anime-specific:** Crucial for sharp anime art style with high-contrast edges

1.2 Interactive Editing

- **GaussianEditor** (Readiness: 9/10)
- **What:** Real-time scene editing (2-7 mins on GPU)
- **Why:** Let users customize generated worlds
- **Implementation:** Fork `GaussianEditor`, replace InstructPix2Pix with anime-tuned diffusion model
- **Impact:** Users can add/remove/modify objects in real-time
- **Integration:** WebUI for browser-based editing before entering world

1.3 Material & Lighting

- **Splatting-Based Inverse Rendering** (Readiness: 8/10)
- **What:** Decompose materials for relighting/editing
- **Why:** Enable anime-style cel shading, dynamic lighting
- **Implementation:** Extend 3DGS with BRDF parameters, use IID + diffusion priors
- **Impact:** Real-time anime shader effects (toon shading, rim lighting)

Deliverables:

- Alias-free 3DGS rendering at 60+ FPS
- User-editable 3D scenes via web interface
- Basic anime-style material system

Phase 2: Dynamic Worlds (3-6 months)

Goal: Transition from static snapshots to animated, living environments

Technologies from Your Research:

2.1 Temporal Dynamics

- **4D Gaussian Splatting** (Readiness: 8/10)
- **What:** Add time dimension to 3DGS for animation
- **Why:** Enable moving water, swaying trees, dynamic scenes
- **Implementation:** Fork `hustv1/4DGaussians`, train on anime video sequences

- **Impact:** Scenes feel alive, not frozen
- **Anime-specific:** Capture iconic anime motions (hair flowing, fabric movement)

2.2 Unbounded Environments

- **SceneDreamer** (Readiness: 8/10)
- **What:** Generate infinite 3D landscapes from 2D images
- **Why:** Users can explore beyond initial frame
- **Implementation:**
 - Train BEV (bird's-eye-view) generator on anime backgrounds
 - Use generative hash grid for scalable terrain
 - Stream content based on user position
- **Impact:** Explore endless anime worlds (Ghibli landscapes, cyberpunk cities)

2.3 Real-Time SLAM

- **Gaussian Splatting SLAM** (Readiness: 9/10)
- **What:** Real-time 3D reconstruction (769 FPS)
- **Why:** Build worlds as users explore, adapt to movement
- **Implementation:** Integrate analytical Jacobian for robust tracking
- **Impact:** Dynamic world expansion, responsive to user exploration

Deliverables:

- Animated 3DGS scenes (wind, water, ambient motion)
- Procedurally extended environments (no boundaries)
- Real-time world reconstruction as users move

Phase 3: Character Integration (6-9 months)

Goal: Populate worlds with interactive, animated characters

Technologies from Your Research:

3.1 Realistic Avatars

- **GaussianAvatars** (Readiness: 8/10)
- **What:** Rigged 3D Gaussian avatars with pose/expression control
- **Why:** Create controllable anime characters
- **Implementation:**
 - Train on anime character datasets (VTuber models, anime renders)
 - Rig to SMPL/FLAME parametric models for animation
 - Drive via user input or AI motion
- **Impact:** Photorealistic anime characters users can interact with
- **Anime-specific:** Adapt for stylized proportions (large eyes, small nose)

3.2 Character Generation

- **AnimeDiffusion** (Readiness: 7/10)
- **What:** Text → anime-style 3D characters with multi-view consistency
- **Why:** Users describe characters, system generates them
- **Implementation:** Fine-tune diffusion model on anime datasets with LoRAs
- **Impact:** On-demand character creation
- **Example:** "Create a pink-haired samurai girl" → instant 3D character

3.3 Real-Time Animation

- **Real-Time Neural Character Animation** (Readiness: 7/10)
- **What:** Online learning for adaptive character responses
- **Why:** Characters react naturally to user actions
- **Implementation:** Integrate with Unity/Unreal, train on interaction datasets
- **Impact:** Characters feel alive, not scripted

Deliverables:

- Library of rigged anime characters
- Text-to-character generation pipeline
- Real-time character animation system

Phase 4: Interaction & Physics (9-12 months)

Goal: Enable users to physically interact with the world

Technologies from Your Research:

4.1 Physics Integration

- **Material-informed Gaussian Splatting** (Readiness: 8/10)
- **What:** Extract material properties, assign physics
- **Why:** Objects behave realistically (bounce, break, collide)
Implementation:
 - Segment 3DGS into material types
 - Assign physics properties (wood, metal, fabric)
 - Integrate with physics engine (Unity/Unreal)
- **Impact:** Pick up swords, throw objects, realistic interactions

4.2 Embodied Navigation

- **SAGE-3D** (Readiness: 7/10)
- **What:** Physically executable 3DGS for embodied AI
- **Why:** Characters navigate world intelligently
- **Implementation:** Add semantic labels + physical interfaces to 3DGS
- **Impact:** NPCs walk around obstacles, use doors, climb stairs

4.3 Active Reconstruction

- **GauSS-MI** (Readiness: 8/10)
- **What:** Smart view selection for reconstruction
- **Why:** Efficiently fill in missing world details
- **Implementation:** Use mutual information for next-best-view planning
- **Impact:** World detail improves as users explore

Deliverables:

- Physics-enabled 3DGS objects
- Intelligent NPC navigation
- Progressive world detail enhancement

Phase 5: World Intelligence (12-18 months)

Goal: Create a living, breathing world with memory and coherence

Technologies from Your Research:

5.1 World Model

- **WorldGen** (Readiness: 6/10)
- **What:** Hierarchical world models with compositional understanding
- **Why:** World remembers user actions, maintains consistency
- **Implementation:** Train on anime world sequences, model state transitions
- **Impact:** Persistent world that evolves based on user actions

5.2 Multimodal Simulation

- **Multimodal World Simulation with Diffusion Prior** (Readiness: 7/10)
- **What:** Integrates vision, language, action for dynamic worlds
- **Why:** World responds to natural language commands
- **Implementation:**
 - Train diffusion model on multi-view anime video
 - Condition on text/action inputs
 - Use for predictive simulation
- **Impact:** "Make it rain" → weather changes; "Summon dragon" → dragon appears

5.3 Large-Scale Streaming

- **Interactive Simulation of Large-Scale Neural Fields** (Readiness: 7/10)
- **What:** Dynamic LOD + streaming for massive worlds
- **Why:** Support city-scale environments without memory limits

- **Implementation:** GPU-accelerated kernel regression, spatial hashing
- **Impact:** Explore entire anime cities (Akihabara, cyberpunk Tokyo)

Deliverables:

- Persistent world with memory
- Natural language interaction
- City-scale environment support

Phase 6: Social & Multiplayer (18-24 months)

Goal: Transform from single-player to shared social experience

Key Requirements (Not in current research - needs R&D):

6.1 Multiplayer Synchronization

- **Challenge:** Multiple users in same 3DGS world
- **Approach:**
 - Centralized server for world state
 - Client-side prediction for low latency
 - Sparse updates for Gaussian parameters
- **Technologies:** WebRTC for networking, distributed rendering

6.2 Social AI Characters

- **Challenge:** NPCs that remember conversations, build relationships
- **Approach:**
 - Integrate LLM (GPT-4o) for dialogue
 - Memory systems (vector DB for conversation history)
 - Emotion models (facial expressions via GaussianAvatars)
- **Technologies:** RAG for character memories, sentiment analysis

6.3 User-Generated Content

- **Challenge:** Users create and share custom worlds/characters

- **Approach:**
- Mod tools (import custom 2D art → 3DGS)
- Asset marketplace
- Community curation system
- **Technologies:** Blockchain for ownership (optional), CDN for distribution

Deliverables:

- Multiplayer support (2-50 players per world)
- AI NPCs with memory and personality
- User content creation tools

Technology Stack Recommendations

Rendering Engine:

- **Primary:** Unity or Unreal Engine 5
- **3DGS Plugin:** gsp1at (nerfstudio-project)
- **Why:** Production-ready game engines with VR/AR support

Backend:

- **Scene Generation:** Modal (GPU inference) + FastAPI
- **World State:** Redis for real-time state, PostgreSQL for persistence
- **Asset Storage:** Vercel Blob or S3
- **Networking:** Photon or custom WebRTC

AI Models:

- **3DGS:** Original INRIA implementation + Mip-Splatting
- **4DGS:** hustvl/4DGaussians
- **Characters:** GaussianAvatars + AnimeDiffusion (fine-tuned)
- **World Gen:** SceneDreamer + DreamGaussian
- **Dialogue:** GPT-4o + custom fine-tuning for anime personalities

Development Priority:

1. **Core Rendering** (3DGS + Mip-Splatting) - 3 months
2. **Interactive Editing** (GaussianEditor) - 2 months
3. **Character System** (GaussianAvatars) - 4 months
4. **World Dynamics** (4DGS + SceneDreamer) - 4 months
5. **Physics & Interaction** (Material-informed GS) - 3 months
6. **World Intelligence** (World Models + Diffusion) - 5 months
7. **Multiplayer** (Custom networking) - 4 months

Total Development Time: 2 years to MVP, 3-4 years to SAO-level experience

Critical Research Gaps (Areas Needing More R&D)

1. Anime Style Preservation

- **Problem:** Most papers focus on photorealism
- **Solution:** Fine-tune on anime-specific datasets (Danbooru, Safebooru)
- **Action:** Create training pipeline for anime 3DGS

2. Real-Time Constraints

- **Problem:** Many methods too slow for 60 FPS VR
- **Solution:** Aggressive optimization, model distillation
- **Action:** Profile every component, optimize bottlenecks

3. Semantic Understanding

- **Problem:** Current 3DGS lacks object semantics
- **Solution:** Integrate SAM2 for segmentation, add semantic labels
- **Action:** Build anime-specific object detector

4. Long-Term Coherence

- **Problem:** World consistency over extended sessions

- **Solution:** Hierarchical world model with checkpointing
- **Action:** Research persistent world state management

5. Interaction Realism

- **Problem:** Physics + 3DGS integration not well-solved
- **Solution:** Hybrid mesh + Gaussian approach
- **Action:** Prototype physics integration with Unity

Recommended Immediate Actions

Month 1-3: Foundation

1. **Integrate Mip-Splatting** into Veloven
2. Fork repo: [niu jinshuchong/mip-splatting](#)
3. Add 3D smoothing filter to training

Test on anime images

Setup GaussianEditor

6. Fork repo: [buaacyw/GaussianEditor](#)
7. Fine-tune InstructPix2Pix on anime dataset

Build web demo for scene editing

Anime Training Pipeline

10. Scrape Danbooru/Safebooru (1M+ anime images)
11. Preprocess: background removal, multi-view generation
12. Train baseline 3DGS on anime data

Month 4-6: Dynamics

1. **4DGS Prototype**
2. Fork repo: [hustvl/4DGaussians](#)
3. Train on anime video clips (Makoto Shinkai films, KyoAni works)

Demo: Flowing hair, moving clouds

SceneDreameer Integration

6. Fork repo: [FrozenBurning/SceneDreameer](#)
7. Train on anime backgrounds
8. Generate unbounded worlds from single frame

Month 7-9: Characters

1. GaussianAvatars for Anime

2. Fork repo: [ShenhanQian/GaussianAvatars](#)
3. Collect anime character dataset (VRoid, MMD models)

Train rigged anime avatars

AnimeDiffusion Pipeline

6. Fine-tune Stable Diffusion with anime LoRAs
7. Implement text → multi-view → 3DGS pipeline
8. Character generation API

Month 10-12: MVP

1. Unity Integration

2. Import 3DGS scenes into Unity
3. Add character controllers

Build VR demo with Meta Quest

Physics Prototype

6. Extract material masks from 3DGS
7. Add Unity physics to objects

Test sword-fighting mechanics

User Testing

- Recruit 10-20 anime fans
 - Test core loop: upload image → explore world → interact with characters
 - Iterate based on feedback
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Success Metrics

Phase 1 (Production 3DGS):

- ■ 60+ FPS at 1080p on RTX 3080
- ■ Alias-free rendering (PSNR > 30 dB)
- ■ Scene editing < 5 minutes

Phase 2 (Dynamic Worlds):

- ■ Animated elements (water, trees) at 60 FPS
- ■ Unbounded exploration (>1km² worlds)
- ■ Real-time SLAM reconstruction

Phase 3 (Characters):

- ■ 10+ rigged anime characters
- ■ Text-to-character < 2 minutes
- ■ Natural character animation

Phase 4 (Interaction):

- ■ Physics-based object interaction
- ■ NPC navigation in complex scenes
- ■ Collision detection accuracy >95%

Phase 5 (Intelligence):

- ■ Persistent world state across sessions
- ■ Natural language commands working
- ■ City-scale environments (10+ km²)

Phase 6 (Social):

- ■ 10+ concurrent users per world
- ■ AI NPCs with personality

- ■ User-generated content tools

Cost Estimates

Development:

- **Team:** 3-5 engineers (2 graphics, 1 ML, 1 fullstack, 1 game dev)
- **Salaries:** \$500K-800K/year
- **Total 2 years:** \$1M-1.6M

Infrastructure:

- **GPU Training:** Modal/RunPod - \$2K-5K/month
- **Inference:** Modal - \$0.015 per generation → scales with users
- **Storage:** Vercel Blob - \$50-500/month
- **Servers:** AWS/GCP - \$500-2K/month
- **Total Year 1:** \$50K-100K

Grand Total (2-year MVP): \$1.1M - \$1.8M

Competitive Landscape

Existing Attempts:

1. **VRChat** - Social VR, but traditional 3D models
2. **Neos VR** - User-generated worlds, no AI generation
3. **Rec Room** - Simple graphics, social focus
4. **Somnium Space** - Metaverse, lacks anime aesthetic
5. **Waifusion** - AI anime generation, not 3D/interactive

Your Advantage:

- **First-mover** in AI-generated anime 3D worlds
- **Technical edge** via 3DGS (faster than NeRF)

- **Anime focus** (untapped niche in VR)
 - **Research-backed** (using cutting-edge papers)
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Risks & Mitigation

Technical Risks:

1. **3DGS too slow for VR**
2. Mitigation: Aggressive LOD, model compression
3. **Anime style loss in 3D**
4. Mitigation: Anime-specific training, toon shaders
5. **World inconsistency**
6. Mitigation: World model checkpointing, validation

Business Risks:

1. **Market too niche**
2. Mitigation: Start with VTuber community, expand
3. **High compute costs**
4. Mitigation: User-pays model, tiered pricing
5. **Content moderation**
6. Mitigation: Automated filters, community reporting

Legal Risks:

1. **Copyright (anime IP)**
 2. Mitigation: User-generated content, fair use
 3. **User safety (VR)**
 4. Mitigation: Safety boundaries, parental controls
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Funding Strategy

Pre-Seed (\$200K-500K):

- **Goal:** Build Phase 1-2 MVP (static → dynamic)
- **Sources:** Angels, anime VCs (Anime Capital, Makers Fund)
- **Pitch:** "Midjourney for anime 3D worlds"

Seed (\$2M-5M):

- **Goal:** Characters + interaction (Phase 3-4)
- **Sources:** a16z, Paradigm (if crypto), game-focused VCs
- **Pitch:** "SAO-like interactive anime worlds"

Series A (\$10M-20M):

- **Goal:** World intelligence + multiplayer (Phase 5-6)
- **Sources:** Traditional VCs, strategic (Sony, Bandai Namco)
- **Pitch:** "The metaverse for anime fans"

Summary: Your Path Forward

What You Have:

■ Working 2D → 3DGS converter (Velowen) ■ Comprehensive research collection ■ Clear technical understanding

What You Need:

■■ Production-ready 3DGS with anime quality ■■ Character animation system ■■ World dynamics and physics
■■ Team and funding

Next 30 Days:

1. **Week 1:** Integrate Mip-Splatting anti-aliasing
2. **Week 2:** Setup GaussianEditor for scene editing
3. **Week 3:** Build anime training dataset (10K images)
4. **Week 4:** Train first anime-specific 3DGS model

Next 90 Days:

- Complete Phase 1 (Production 3DGS)
- Prototype Phase 2 (4DGS animation)
- Recruit first engineer/co-founder
- Start fundraising conversations

Your SAO Dream is Achievable in 2-3 Years

The technology exists. Your research is solid. The market is ready (VTubers, anime VR fans, metaverse hype).

The only thing between you and Sword Art Online is execution.

Start with Mip-Splatting this week. Build one phase at a time. In 2 years, you'll have users exploring anime worlds they created from a single image.

Let's build the future of anime together. ■■■

Additional Resources

Papers to Read Immediately:

1. Mip-Splatting (CVPR 2024 Best Student Paper)
2. GaussianAvatars (for characters)
3. 4D Gaussian Splatting (for dynamics)
4. Gaussian Splatting SLAM (for real-time)

GitHub Repos to Fork:

1. graphdeco-inria/gaussian-splatting
2. niujinshuchong/mip-splatting
3. buaacyw/GaussianEditor
4. ShenhanQian/GaussianAvatars
5. hustvl/4DGaussians

Communities to Join:

1. 3DGS Discord (graphdeco-inria)

2. Nerfstudio Discord
3. VRChat Creators Discord
4. /r/anime and /r/virtualreality
5. Twitter: Follow authors of key papers

Datasets to Collect:

1. Danbooru (anime images)
2. VRoid Hub (anime characters)
3. MMD models (character rigging)
4. Anime backgrounds (Shinkai, Ghibli style)
5. Anime video clips (for 4DGS training)

This roadmap is based on your January 7, 2026 research compilation. As new papers emerge, update this document.

Good luck building the anime metaverse! ■